Fall 2024 CS4641/CS7641 Homework 1

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Deadline: Friday, September 20th, 11:59 pm EST

- No unapproved extension of the deadline is allowed. For late submissions, please refer to the course website.
- Discussion is encouraged on Ed as part of the Q/A. However, all assignments should be done individually.
- Plagiarism is a **serious offense**. You are responsible for completing your own work. You are not allowed to copy and paste, or paraphrase, or submit materials created or published by others, as if you created the materials. All materials submitted must be your own. This also means you may not submit work created by generative models as your own.
- All incidents of suspected dishonesty, plagiarism, or violations of the Georgia Tech Honor Code will be subject to the institute's Academic Integrity procedures. If we observe any (even small) similarities/plagiarisms detected by Gradescope or our TAs, **WE WILL DIRECTLY REPORT ALL CASES TO OSI**, which may, unfortunately, lead to a very harsh outcome. **Consequences can be severe, e.g., academic probation or dismissal, grade penalties, a 0 grade for assignments concerned, and prohibition from withdrawing from the class**.

Instructions

- We will be using Gradescope for submission and grading of assignments.
- **Unless a question explicitly states that no work is required to be shown, you must provide an explanation, justification, or calculation for your answer.** Basic arithmetic can be combined (it does not need to each have its own step); your work should be at a level of detail that a TA can follow it.
- Your write-up must be submitted in PDF form, you may use either Latex, markdown, or any word processing software. We will **NOT** accept handwritten work. Make sure that your work is formatted correctly, for example submit $\sum_{i=0} x_i$ instead of sum_{i=0} x_i.
- **A useful video tutorial on LaTeX has been created by our TA team** and can be found here and an Overleaf document with the commands can be found here.
- When submitting your assignment on Gradescope, **you are required to correctly map pages of your PDF to each question/ subquestion to reflect where they appear.** Improperly mapped questions will not be graded correctly.
- All assignments should be done individually, each student must write up and submit their own answers.
- **Graduate Students**: You are required to complete any sections marked as Bonus for Undergrads

*Point Distribution

Q1: Linear Algebra [28pts]

- 1.1 Determinant and Inverse of a Matrix [10pts]
- 1.2 Eigenvalues and Eigenvectors [20pts]

Q2: Expectation, Co-variance and Statistical Independence [7pts]

Q3: Optimization [17pts: 17pts + 2% Bonus for All]

Q4: Maximum Likelihood [20pts: 10pts + 10 pts Grad/6% Bonus for Undergrads]

- 4.1 Discrete Example [10pts]
- 4.2 Poisson Distribution [10pts Grad / 6% Bonus for Undergrads]

Q5: Information Theory [26pts]

- 5.1 Mutual Information and Entropy [16pts]
- 5.2 Entropy Proofs [10pts]

Q6: Ethical Implications on Decision-Making [5 pts]

Q7: Programming [5pts]

Q8: Bonus for All [8%]

Points Totals:

- **Total Base:** 100 pts
- **Total Undergrad Bonus:** 6%
- **Total Bonus for All:** 10%
- **Total Possible Assignment Grade (Undergrad):** 116%
- **Total Possible Assignment Grade (Grad):** 110%

1 Linear Algebra [10pts + 18pts]

1.1 Determinant and Inverse of Matrix [10pts]

Given a matrix M:

$$M = \begin{bmatrix} 3 & 1 & 4 \\ r & 2 & -4 \\ 0 & -3 & 5 \end{bmatrix}$$

(a) Calculate the determinant of $\boldsymbol{M}$ in terms of r (calculation process is required). [4pts]

$$|\boldsymbol{M}| = 3\left|\begin{bmatrix} 2 & -4 \\ -3 & 5 \end{bmatrix}\right| - r\left|\begin{bmatrix} 1 & 4 \\ -3 & 5 \end{bmatrix}\right| + 0\left|\begin{bmatrix} 1 & 4 \\ 2 & -4 \end{bmatrix}\right|$$

$$|\boldsymbol{M}| = 3((2)(5) - (-4)(-3)) - r((1)(5) - (-3)(4)) + 0((1)(-4) - (2)(4))$$

$$|\boldsymbol{M}| = -6 - 17r + 0$$

$$|\boldsymbol{M}| = -6 - 17r$$

(b) For what value(s) of r does $\boldsymbol{M}^{-1}$ not exist? Why doesn't $\boldsymbol{M}^{-1}$ exist in this case? What does it mean in terms of rank and singularity for these values of r? *This question can be answered in less than 7 lines.* [3pts]

We know M^{-1} does not exist if $|\boldsymbol{M}| = 0$. Therefore, we find $|\boldsymbol{M}| = 0$ when $-6 - 17r = 0$. We can easily see when $r = -\frac{6}{17}$, $|\boldsymbol{M}| = 0$.

We know for $\mathbf{M}$ to have an inverse, $\mathbf{M}$ must be full rank. For $r = -\frac{6}{17}$, it is evident, by definition, $\mathbf{M}$ is no longer full rank and thus is considered singular or non-inevitable since an inverse does not exist.

(c) Find the mathematical equation that describes the relationship between the determinant of $\boldsymbol{M}$ and the determinant of $\boldsymbol{M}^{-1}$. [3pts]

NOTE: It may be helpful to find the determinant of $\boldsymbol{M}$ and $\boldsymbol{M}^{-1}$ for $r = 0$.

$$|\mathbf{M}^{-1}| = \frac{1}{|\mathbf{M}|}$$

1.2 Eigenvalues and Eigenvectors [5+15pts]

1.2.1 Eigenvalues [5pts]

Given the following matrix $\boldsymbol{A}$, find an expression for the eigenvalues λ of $\boldsymbol{A}$ in terms of a, b, and c. [5pts]

$$\boldsymbol{A} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} a - \lambda & b \\ b & c - \lambda \end{bmatrix}$$

$$|A - \lambda I| = (a - \lambda)(c - \lambda) - (b)(b)$$

$$|A - \lambda I| = ac - a\lambda - c\lambda + \lambda^2 - b^2$$

$$ac - a\lambda - c\lambda + \lambda^2 - b^2 = 0$$

$$\lambda^2 - (a + c)\lambda + (ac - b^2) = 0$$

Using the Quadratic Equation, we find:

$$\lambda = \frac{(a + c) \pm \sqrt{(a - c)^2 + 4b^2}}{2}$$

1.2.2 Eigenvectors [15pts]

Given a matrix $\boldsymbol{A}$:

$$\boldsymbol{A} = \begin{bmatrix} 11 & 4 \\ 4 & 5 \end{bmatrix}$$

(a) Calculate the eigenvalues of $\boldsymbol{A}$. [3pts]

$$A - \lambda I = \begin{bmatrix} 11 - \lambda & 4 \\ 4 & 5 - \lambda \end{bmatrix}$$

$$|A - \lambda I| = (11 - \lambda)(5 - \lambda) - (4)(4)$$

$$|A - \lambda I| = 55 - 11\lambda - 5\lambda + \lambda^2 - 16$$

$$\lambda^2 - 16\lambda + 39 = 0$$

Using the Quadratic Equation, we find:

$$\lambda = \frac{16 \pm \sqrt{100}}{2}$$

Therefore, we find $\lambda = 3$ and $\lambda = 13$.

(b) Find the normalized eigenvectors of matrix $\boldsymbol{A}$ (calculation process required). [7pts]

With eigenvalues of $\lambda = 3$ and $\lambda = 13$, we will find the respective eigenvectors in two parts.

For $\lambda = 3$:

$$A - \lambda I = A - 3I = \begin{bmatrix} 11 - 3 & 4 \\ 4 & 5 - 3 \end{bmatrix} = \begin{bmatrix} 8 & 4 \\ 4 & 2 \end{bmatrix}$$

$$8x_1 + 4x_2 = 0 \tag{1}$$
$$4x_1 + 2x_2 = 0 \tag{2}$$

$$x_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

To normalize this vector, we find $\sqrt{1^2 + (-2)^2} = \sqrt{5}$. Therefore, as the normalized eigenvector, we find:

$$x_2 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$$

For $\lambda = 13$:

$$A - \lambda I = A - 13I = \begin{bmatrix} 11 - 13 & 4 \\ 4 & 5 - 13 \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ 4 & -8 \end{bmatrix}$$

$$-2x_1 + 4x_2 = 0 \tag{3}$$
$$4x_1 + -8x_2 = 0 \tag{4}$$

$$x_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

To normalize this vector, we find $\sqrt{2^2 + 1^2} = \sqrt{5}$. Therefore, as the normalized eigenvector, we find:

$$x_2 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$$

(c) If done correctly, the normalized eigenvectors from part (b) and the matrix $(\boldsymbol{A} - \lambda \boldsymbol{I})$ are both nonzero. Despite both being nonzero, we still have $(\boldsymbol{A} - \lambda \boldsymbol{I})x = 0$ (where x is an eigenvector). What are some properties of the matrix $(\boldsymbol{A} - \lambda \boldsymbol{I})$ which allow for this? Additionally, why is the determinant $|\boldsymbol{A} - \lambda \boldsymbol{I}| = 0$? [5pts]

NOTE: There are many ways to solve this problem. You are allowed to use linear algebra properties as part of your solution.

If λ is an eigenvalue of **A**, then there exists a non-zero vector x (an eigenvector) such that $(A - \lambda I)x = 0$. Geometrically, this means that **A** scales the vector x by λ, which implies that x is in the direction of the eigenvector and its magnitude is scaled by λ. This implies that the matrix $(A - \lambda I$ is singular, which means it does not have full rank. Therefore, there exists a non-zero solution for x. Since $(A - \lambda I) = 0$, we know $(A - \lambda I)$ has a null space, which corresponds to the eigenspace of **A**.

There exist two properties of $(A - \lambda I)$ that allow $(A - \lambda I) = 0$ to be true.

(a) The matrix $(A - \lambda I)$ is singular. By definition, the determinant must be zero.

(b) $(A - \lambda I) = 0$ being true, as previously established, implies there exists a non-trivial $(x \neq 0)$ null-space for the matrix $(A - \lambda I)$. Using what was previously defined, we know the eigenvector lies in the null space.

Furthermore, the determinant is zero since the matrix $(A - \lambda I)$ is singular. This occurs since λ is an eigenvalue of **A**, whose eigenvector exists in the null space.

2 Expectation, Co-variance and Statistical Independence [7pts]

Suppose X, Y, and Z are three different random variables. Let X obey a two point Distribution. The probability mass function for X is:

$$p(x) = \begin{cases} 0.9 & x = c \\ 0.1 & x = -c \end{cases}$$

where c is a nonzero constant. The distribution of Y is not known, but it is provided $Var(Y) = 1.44c^2$. X and Y are statistically independent (i.e. $P(X|Y) = P(X)$). Meanwhile, let $Z = 4X + 2Y$.

Calculate the correlation coefficient defined as $\rho(X, Z) = \frac{Cov(X,Z)}{\sqrt{Var(X)Var(Z)}}$. Round your answer to 3 decimal places or simplified radical form.

HINT: Review the probability and statistics lecture slides

We begin by calculating $Cov(X, Z)$. Steps are shown below, utilizing the distributive property, the ability to pull out constants, and the variance-covariance relationship.

$$Cov(X, Z) = Cov(X, 4X + 2Y) = Cov(X, 4X) + Cov(X + 2Y)$$

$$Cov(X, 4X + 2Y) = 4Cov(X, X) + 2Cov(X, Y)$$

$$Cov(X, 4X + 2Y) = 4Var(X) + 2Cov(X, Y)$$

Since X and Y are statistically independent, we find relatively trivially $2Cov(X, Y) = 0$. Therefore, we arrive at our answer of $Cov(X, Z) = 4Var(X)$.

We will then calculate $Var(X)$, since this is the next logical step and not because I realized the following step requires completion of this step. Steps are shown below.

$$Var(X) = E[X^2] - E[X]^2$$

$$E[X] = 0.9(c) + 0.1(-c) = 0.9c - 0.1c = 0.8c$$

$$E[X]^2 = 0.9(c)^2 + 0.1(c)^2 = 0.9c^2 + 0.1c^2 = c^2$$

$$Var(X) = E[X^2] - E[X]^2 = c^2 - (0.8c)^2 = c^2 - 0.64c^2 = 0.36c^2$$

We will then calculate $Var(Z)$, whose steps are shown below, utilizing the distributive property and the ability to pull out constants (while squaring) - as seen before.

$$Var(Z) = Var(4X + 2Y) = 4^2Var(X) + 2^2Var(Y)$$

$$Var(Z) = 16(0.36c^2) + 4(1.44c^2) = 11.52c^2$$

A simple plug-and-chug will yield the following:

$$\rho(X, Z) = \frac{Cov(X, Z)}{\sqrt{Var(X)Var(Z)}} = \frac{4(0.36c^2)}{\sqrt{0.36c^2 \times 11.52c^2}} = \frac{1.44}{\sqrt{4.1472}} \approx 0.707$$

3 Optimization [17pts + 2% Bonus for All]

Optimization problems are related to minimizing a function (usually termed loss, cost or error function) or maximizing a function (such as the likelihood) with respect to some variable x. The Karush-Kuhn-Tucker (KKT) conditions are first-order conditions for a solution in nonlinear programming to be optimal, provided that some regularity conditions are satisfied. In this question, you will be solving the following optimization problem:

$$\begin{aligned}
\max_{x,y} \quad & f(x,y) = 2x + 3xy \\
\text{s.t.} \quad & g_1(x,y) = x^2 + 4y^2 \leq 9 \\
& g_2(x,y) = y \leq \frac{1}{2}
\end{aligned}$$

(a) Write the Lagrange function for the maximization problem. Now change the maximum function to a minimum function (i.e. $\underset{x,y}{min} \ f(x,y) = 2x+3xy$) and provide the Lagrange function for the minimization problem with the same constraints g_1 and g_2. [2pts]

NOTE: The minimization problem is only for part (a).

We define the Lagrange function to be $L(x, y, \lambda_1, \lambda_2) = f(x, y) - \lambda_1 g_1(x, y) - \lambda_2 g_2(x, y)$.

We then will isolate the constraint function $g_1(x, y)$ to be $x^2 + 4y^2 - 9$ and $g_2(x, y)$ to be $y - \frac{1}{2}$.

Therefore, we arrive at our maximization of $L(x, y, \lambda_1, \lambda_2) = 2x + 3xy - \lambda_1(x^2 + 4y^2 - 9) - \lambda_2(y - \frac{1}{2})$

Similarly, we arrive at our minimization of $L(x, y, \lambda_1, \lambda_2) = 2x + 3xy + \lambda_1(x^2 + 4y^2 - 9) + \lambda_2(y - \frac{1}{2})$

As a note for above, we wish to follow a simple set of rules for inequality constraints:

(a) Elements in the constraint function should be arranged such that $g(x) = x \leq 0$, where x may be constants and/or variables.

(b) For minimization problems, the Lagrange Multiplier component should be positive. For maximization problems, the Lagrange Multiplier component should be negative.

(b) List the names of all 4 groups of KKT conditions and their corresponding mathematical equations or inequalities for this specific maximization problem. [2pts]

(a) No Feasible Descent (Stationary): There is no feasible direction that could improve the objective function.

$$\nabla f(X) - \sum \lambda \nabla g_i(x) = 0$$

(b) Primal Feasibility: The constraints must not be violated by the optimal conditions.

$$h_j(x) = 0$$
$$g_i(x) \leq 0$$

(c) Complementary Slackness (Complementary): The product of the Lagrange Multipliers and the corresponding variables must be zero.

$$\sum \lambda_i g_i(x) = 0$$

(d) Dual Feasibility: The Lagrange Multipliers associated with the constraints must be non-negative.

$$\lambda_i \geq 0$$

(c) Solve for 4 possibilities formed by each constraint being active or inactive. Do not forget to check the inactive constraints for each point when applicable. Candidate points must satisfy all the conditions mentioned in part b). [8pts]

We begin by taking the derivative of the Maximization equation with respect to x and y as well as defining the slack constraints.

$$\frac{dL}{dx} = 2 + 3y - 2\lambda_1 x = 0$$
$$\frac{dL}{dy} = 3x - 8\lambda_1 y - \lambda_2 = 0$$
$$\lambda_1(x^2 + 4y^2 - 9) = 0$$
$$\lambda_2(y - \frac{1}{2}) = 0$$

Next, we will calculate the candidates for Active, Active.

We first define the equations we will be working with:

$$x^2 + 4y^2 - 9 = 0, \ where \ \lambda_1 > 0$$

$$y - \frac{1}{2} = 0, \ where \ \lambda_2 > 0$$

Now, relatively trivially, we will solve for x and y.

$$y = \frac{1}{2}$$

$$x^2 + 4(\frac{1}{2})^2 - 9 = 0$$
$$x = \pm\sqrt{8}$$

We find x has two values. Therefore, we will separate the conditions to find the candidate.

Condition One: $(\sqrt{8}, \frac{1}{2})$

$$2 + \frac{3}{2} - 2\lambda_1\sqrt{8} = 0$$
$$\frac{7}{2} = s\sqrt{8}\lambda_1$$
$$\lambda_1 = \frac{7}{4\sqrt{8}} > 0$$
$$3\sqrt{8} - \frac{4(7)}{4\sqrt{8}} - \lambda_2 = 0$$
$$\lambda_2 = 3\sqrt{8} - \frac{7}{\sqrt{8}} \approx 6.010 > 0$$

We find Condition One follows the KKT conditions and is a candidate.

Condition Two: $(-\sqrt{8}, \frac{1}{2})$

$$2 + \frac{3}{2} + 2\sqrt{8}\lambda_1 = 0$$
$$\frac{7}{2} = -2\sqrt{8}\lambda_1$$
$$\lambda_1 = -\frac{7}{4\sqrt{8}} < 0$$

Therefore, the KKT conditions do not hold for Condition Two.

We will now calculate the candidates for Active, Inactive. We first define the equations we will be working with:

$$x^2 + 4y^2 - 9 = 0, \ where \ \lambda_1 > 0$$

$$y - \frac{1}{2} \leq 0, \ where \ \lambda_2 = 0$$

Solving for x, we obtain $x = \pm\sqrt{9 - 4y^2}$. As such, we again need two conditions.

Condition One: $x = \sqrt{9 - 4y^2}$

$$2 + 3y - 2\lambda_1\sqrt{9 - 4y^2} = 0$$

$$2 + 3y = 2\lambda_1\sqrt{9 - 4y^2}$$

$$\lambda_1 = \frac{2 + 3y}{2\sqrt{9 - 4y^2}}$$

$$2\sqrt{9 - 4y^2} - 8(\frac{2 + 3y}{2\sqrt{9 - 4y^2}})y - 0 = 0$$

$$y = 0.769 \ or \ -1.169$$

Since $y = 0.769 - \frac{1}{2}$ is not less than or equal to 0, we find this value to not be valid. Instead, we will check $y = -1.169$.

$$\lambda_1 = \frac{2 + (3)(-1.169)}{2\sqrt{9 - 4(-1.169)^2}}$$

Without doing much arithmetic, we see the numerator is negative whereas the denominator is positive, resulting in $\lambda_1 < 0$, which makes the value of $y = -1.169$ invalid.

Condition Two: $x = -\sqrt{9 - 4y^2}$

$$2 + 3y + 2\lambda_1\sqrt{9 - 4y^2} = 0$$

$$\lambda_1 = -\frac{2 + 3y}{2\sqrt{9 - 4y^2}}$$

$$-3\sqrt{9 - 4y^2} + 8(\frac{2 + 3y}{2\sqrt{9 - 4y^2}})y - 0 = 0$$

$$y = -1.240 \ or \ 0.907$$

Through similar reason as before, $y = 0.907$ is invalid since $y = 0.907 - \frac{1}{2}$ is not less than or equal to 0. Therefore, we will check the other value.

$$\lambda_1 = -\frac{2 + 3(-1.240)}{2\sqrt{9 - 4(-1.240)^2}} = 0.509 > 0$$

We find Condition One follows the KKT conditions and is a candidate.

We will now calculate the candidates for Inactive, Active. We first define the equations we will be working with:

$$x^2 + 4y^2 - 9 \leq 0, \ where \ \lambda_1 = 0$$

$$y - \frac{1}{2} = 0, \ where \ \lambda_2 > 0$$

Trivially, we have $y = \frac{1}{2}$ and $\lambda_1 = 0$. As such, using the $\frac{dL}{dx} = 2 + 3y - 2\lambda_1 x = 0$, we have $2 + \frac{3}{2} - 0 = 0$, which is not true. Therefore, this candidate is not valid.

We will now calculate the candidates for Inactive, Inactive. We first define the equations we will be working with:

$$x^2 + 4y^2 - 9 \leq 0, \ where \ \lambda_1 = 0$$

$$y - \frac{1}{2} \leq 0, \ where \ \lambda_2 = 0$$

Through substitution with $\frac{dL}{dy}$, we easily achieve the following:

$$3x - 0 - 0 = 0$$

Therefore, $x = 0$. Solving for y, we have:

$$2 + 3y - 0 = 0$$

Therefore, we easily see $(x, y) = (0, -\frac{2}{3})$. We find this candidate holds true for the KKT conditions.

(d) List the candidate point(s) (there is at least 1) obtained from part c). Please round answers to 3 decimal points and use that answer for calculations in further parts. This part can be completed in one line per candidate point. [2pts]

(a) For Active, Active: $(x, y) = (\sqrt{8}, \frac{1}{2}), \ \lambda_1 = \frac{7}{4\sqrt{8}} \approx 0.619, \ \lambda_2 = 3\sqrt{8} - \frac{7}{\sqrt{8}} \approx 6.010$

(b) For Active, Inactive: $(x, y) = (-1.688, -1.240), \ \lambda_1 = 0.509, \ \lambda_2 = 0$

(c) For Inactive, Inactive: $(x, y) = (0, -\frac{2}{3}), \ \lambda_1 = 0, \ \lambda_2 = 0$

(e) Find the **one** candidate point for which $f(x, y)$ is largest. Check if $L(x, y)$ is concave, convex, or neither at this point by using the Hessian in the second partial derivative test. [3pts]

We remember $f(x, y) = 2x + 3xy$. We will calculate with the order in the previous part.

(a) $f(\sqrt{8}, \frac{1}{2}) = 9.899$

(b) $f(-1.688, -1.240) = 2.903$

(c) $f(0, -\frac{2}{3}) = 0$

Therefore, we easily see the one candidate for which $f(x, y)$ is the largest is: Active, Active: $(x, y) = (\sqrt{8}, \frac{1}{2}), \ \lambda_1 = \frac{7}{4\sqrt{8}} \approx 0.619, \ \lambda_2 = 3\sqrt{8} - \frac{7}{\sqrt{8}} \approx 6.010$.

We will now use the Hessian matrix to determine if $L(x, y)$ is concave, convex, or neither at the point.

$$\mathbf{H}f(x, y) = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$$

We remember from the previous part(s):

$$\frac{\partial}{\partial x} = 2 + 3y - 2\lambda_1 x \ and \ \frac{\partial}{\partial y} = 3x - 8\lambda_1 y - \lambda_2$$

Therefore, we find:

$$\frac{\partial^2 f}{\partial x^2} = -2\lambda_1 = -2\lambda_1$$

$$\frac{\partial^2 f}{\partial x \partial y} = 3$$

$$\frac{\partial^2 f}{\partial y \partial x} = 3$$

$$\frac{\partial^2 f}{\partial y^2} = -8\lambda_1 = -8\lambda_1$$

We remember $\lambda_1 = 0.619$. As such, we have the Hessian matrix defined as:

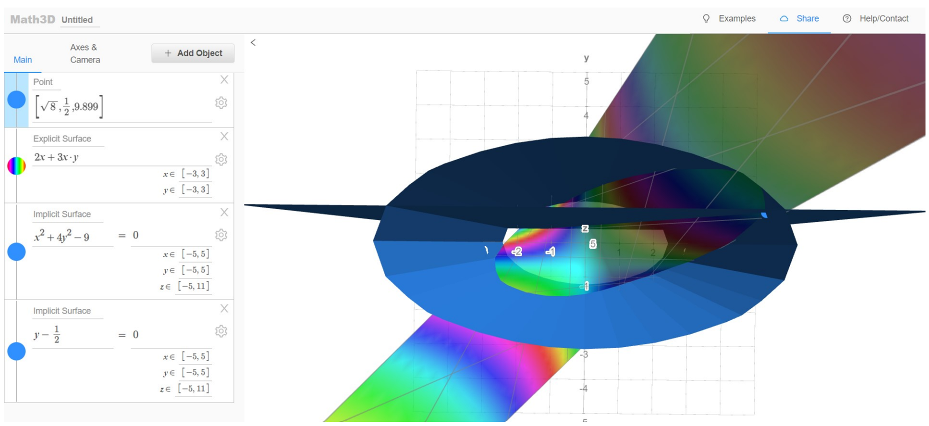
$$\mathbf{H}f(x,y) = \mathbf{H}f(\sqrt{8}, \frac{1}{2}) = \begin{bmatrix} -2(0.619) & 3 \\ 3 & -8(0.619) \end{bmatrix} = \begin{bmatrix} -1.238 & 3 \\ 3 & -4.952 \end{bmatrix}$$

We find $det(\mathbf{H}f(x,y)) = 6.131 - (3)(3) = -2.869$. Therefore, since $det(\mathbf{H}f(x,y)) < 0$, we find $(x,y) = (\sqrt{8}, \frac{1}{2})$ to be a saddle point.

(f) **BONUS FOR ALL:** Make a 3D plot of the objective function $f(x,y)$ and constraints g_1 and g_2 using Math3d. Mark the maximum candidate point and include a screenshot of your plot. Briefly explain why your plot makes sense in one sentence. Although this is bonus, this is **VERY HELPFUL** in understanding what was accomplished in this problem. [2%]

NOTE: Use an explicit surface for the objective function, implicit surfaces for the constraints, and a point for the minimum candidate point.

Through the graph we are able to view correctness since the point $(\sqrt{8}, \frac{1}{2}, 9.899)$ rests at the maximum point on the graph when bounded by the two constraints.
Here's the link to the graph: https://www.math3d.org/LFebM4Cvu.



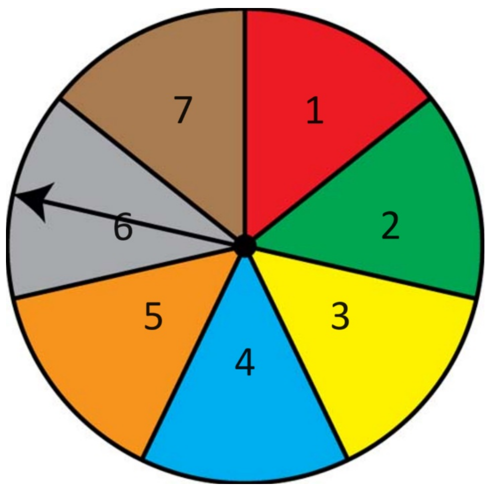
HINT: Read the Example_optimization_problem.pdf in Canvas Files for HW1 to see an example with some explanations.
HINT: Click here for a video explaining the intuition behind KKT problems.
HINT: Click here for an example maximization problem. It's recommended to only watch up until 23:14.
HINT: Click here to determine how to set up the problem for minimization in part (a) and for KKT conditions in part (b).

4 Maximum Likelihood [10pts + 10pts Grads / 6% Bonus for Undergrads]

4.1 Discrete Example [10pts]

Mastermind Mahdi decides to give a challenge to his students for their MLE Final. He provides a spinner with 7 sections, each numbered 1 through 7. The students can change the sizes of each section, meaning that they can select the probability the spinner lands on a certain section. Mahdi then proposes that the students will get a 100 on their final if they can spin the spinner 7 times such that it doesn't land on section 1 during the first 6 spins and lands on section 1 on the 7th spin. If the probability of the spinner landing on section 1 is θ, what value of θ should the students select to most likely ensure they get a 100 on their final? Use your knowledge of Maximum Likelihood Estimation to get a 100 on the final.

NOTE: You must specify the log-likelihood function and use MLE to solve this problem for full credit. You may assume that the log-likelihood function is concave for this question



We note the Bernoulli Objective Function discussed in class is as follows:

$$P(x_i|\theta) = \theta^{x_i}(1-\theta)^{1-x_i}$$

This Objective Function depicts the mean likelihood estimation where failures are represented by the $(1-\theta)^{1-x_i}$ term and success is represented by the θ^{x_i} term.
For the MLE function, we have x = observed data points and θ = unknown parameters.
Using the previously defined information, we have the Objective Function for the problem defined as

$$L(\theta) = (1-\theta)^6\theta$$

It is important to note that since the sequence is fixed (6 Failure, then 1 Success), there is no variability in the position in the success and therefore there is no need to sum over any possibilities.
We will now calculate the log-likelihood function to be

$$\mathcal{L}(\theta) = log(L(\theta)) = log((1-\theta)^6\theta) = 6log(1-\theta) + log(\theta)$$

Next, we will take the derivative with respect to theta to find the critical point.

$$\frac{d\mathcal{L}}{d\theta} = -\frac{6}{1-\theta} + \frac{1}{\theta} = 0$$

Solving for θ, we get $\theta = \frac{1}{7}$. Since we assume the log-likelihood function is concave, we know this value is a maximum. Therefore, we have arrived at our answer of $\theta = \frac{1}{7}$.

4.2 Poisson distribution [10 pts Grad / 6% Bonus for Undergrads]

The Poisson distribution is defined as:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}(k = 0, 1, 2, ...).$$

(a) Let $X_1 \sim Poisson(\lambda)$. What is the likelihood of λ given x_1 is an observed value of X_1?[2 pts / 1%]

Given to us is the Poisson Distribution. We wish to find the likelihood of λ given x_1. We will start with the Probability Mass Function.

$$P(X = x_1) = \frac{\lambda^{x_1} e^{-\lambda}}{x_1!}$$

Now, we wish to convert this into a Likelihood Function.

$$L(\lambda|x_1) = \frac{\lambda^{x_1} e^{-\lambda}}{x_1!}$$

Relatively simply, we have arrived at our answer.

(b) Now, assume we are given n such values. Let $(X_1, ..., X_n) \sim Poisson(\lambda)$ where $X_1, ..., X_n$ are i.i.d. random variables, and $x_1, ..., x_n$ be observed values of $X_1, ..., X_n$. What is the likelihood of λ given this data? You may leave your answer in product form. [2 pts / 1%]

Using the lecture notes as a guide, we know the following:

$$L(\lambda|X) = L(\lambda|X = x_1, X = x_2, ..., X = x_n)$$

We will make an assumption of independent and identically distributed random variables. Therefore, we can simplify to the following:

$$L(\lambda|X) = \Pi_{i=1}^n P(x_i|\lambda) = \Pi_{i=1}^n \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}$$

(c) What is the maximum likelihood estimator of λ? [6 pts / 4%]

We start with simplifying $L(\lambda|X)$ into something more manageable.

$$L(\lambda|X) = (\frac{\lambda^{x_1} e^{-\lambda}}{x_1!})(\frac{\lambda^{x_2} e^{-\lambda}}{x_2!})(...)(\frac{\lambda^{x_n} e^{-\lambda}}{x_n!}) = \frac{\lambda^{\sum x_i} e^{-\lambda}}{\sum x_i!}$$

Now, we will find $\mathcal{L}(\lambda|X) = log(L(\lambda|X))$.

$$\mathcal{L}(\lambda|X) = log(\frac{\lambda^{\sum_{i=1}^n x_i} e^{-n\lambda}}{\Pi x_i!}) \tag{5}$$

$$= log(\lambda^{\sum_{i=1}^n x_i} e^{-n\lambda}) - log(\Pi_{i=1}^n x_i!) \tag{6}$$

$$= \sum_{i=1}^n x_i log(\lambda) + (-n\lambda) log(e) - \sum_{i=1}^n log(x_i!) \tag{7}$$

Since the use of ln or log is arbitrary, we can approximate $log(e) \approx 1$. Therefore, we arrive at our working equation of

$$\mathcal{L}(\lambda|X) = \sum_{i=1}^n x_i log(\lambda) + (-n\lambda) - \sum_{i=1}^n log(x_i!)$$

We will now calculate the derivative with respect to lambda.

$$\frac{d\mathcal{L}}{d\lambda} = \frac{\sum_{i=1}^{n} x_i}{\lambda} - n - 0 = 0$$

We proceed by solving for λ.

$$n = \frac{\sum_{i=1}^{n} x_i}{\lambda}$$

$$\lambda n = \sum_{i=1}^{n} x_i$$

$$\lambda = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Therefore, we find the maximum likelihood estimator of λ is $\hat{\lambda} = \frac{1}{n} \sum_{i=1}^{n} x_i$. This answer is logical since the MLE for λ in a Poisson Distribution is the sample mean of the observed values.

5 Information Theory [16pts + 10pts]

5.1 Mutual Information and Entropy [16pts]

A recent study has shown symptomatic infections are responsible for higher transmission rates. Using the data collected from positively tested patients, we wish to determine which feature(s) have the greatest impact on whether or not some will present with symptoms. To do this, we will compute the entropies, conditional entropies, and mutual information of select features. Please use base 2 when computing logarithms.

ID	**Vaccine Doses** (X_1)	**Wears Mask?** (X_2)	**Underlying Conditions** (X_3)	**Symptomatic** (Y)
1	L	T	F	F
2	L	F	T	T
3	L	F	F	F
4	H	T	F	F
5	L	F	T	T
6	H	F	T	T
7	L	F	T	F
8	M	F	F	T
9	H	T	F	T
10	M	T	F	F

Table 1: Vaccine Doses: {(H) booster, (M) 2 doses, (L) 1 dose, (T) True, (F) False}

(a) Find entropy $H(Y)$ to at least 3 decimal places. [3pts]

To calculate the entropy for $H(Y)$, we must first count the number of times a data point is symptomatic or is asymptomatic. We find $symptomatic = 5$ and $asymptomatic = 5$.

We will now define entropy to be $H(Y) = -\sum_{k=1}^{k} P(Y = k) log_2 P(Y = k)$. Since there are 10 data points, we find $P(Y = F) = P(Y = T) = \frac{5}{10}$. As such, we define $H(Y)$ as follows:

$$H(Y) = -(\frac{5}{10} log_2(\frac{5}{10}) + \frac{5}{10} log_2(\frac{5}{10})) = -(-\frac{1}{2} - \frac{1}{2}) = 1$$

Our answer of $H(Y) = 1$ makes sense here because there are 10 data points with $P(Y = F) = P(Y = T) = \frac{1}{2}$. This is no different, statistically, than flipping a fair coin and calculating the entropy - which is also 1.

(b) Find the average conditional entropy $H(Y|X_1)$ and $H(Y|X_2)$ to at least 3 decimal places. [7pts]

We recognize we will be calculating the conditional entropy for a discrete variable, which is defined as:

$$H(Y|X_1) = \sum_{x \in X_1, y \in Y} p(x_i, y_i) log_2(\frac{p(x_i)}{p(x_i, y_i)})$$

We have three cases for X_1: $X_1 = L$, $X_1 = M$, $X_1 = H$ and two cases for Y: $Y = F$, $Y = T$. Therefore, we find the following:

$$
\begin{aligned}
H(Y|X_1) &= p(L,F)\log_2\left(\frac{p(L)}{p(L,F)}\right) + p(L,T)\log_2\left(\frac{p(L)}{p(L,T)}\right) \\
&\quad + p(M,F)\log_2\left(\frac{p(M)}{p(M,F)}\right) + p(M,T)\log_2\left(\frac{p(M)}{p(M,T)}\right) \\
&\quad + p(H,F)\log_2\left(\frac{p(H)}{p(H,F)}\right) + p(H,T)\log_2\left(\frac{p(H)}{p(H,T)}\right) \\
&= \frac{3}{10}\log_2\left(\frac{\frac{5}{10}}{\frac{3}{10}}\right) + \frac{2}{10}\log_2\left(\frac{\frac{5}{10}}{\frac{2}{10}}\right) \\
&\quad + \frac{1}{10}\log_2\left(\frac{\frac{2}{10}}{\frac{1}{10}}\right) + \frac{1}{10}\log_2\left(\frac{\frac{2}{10}}{\frac{1}{10}}\right) \\
&\quad + \frac{1}{10}\log_2\left(\frac{\frac{3}{10}}{\frac{1}{10}}\right) + \frac{2}{10}\log_2\left(\frac{\frac{3}{10}}{\frac{2}{10}}\right) \\
&= 0.960960
\end{aligned}
$$

Using similar logic to above, we have two cases for X_2: $X_2 = F,\ X_2 = T$ and two cases for Y: $Y = F,\ Y = T$.

$$
\begin{aligned}
H(Y|X_2) &= p(F,F)\log_2\left(\frac{p(F)}{p(F,F)}\right) + p(F,T)\log_2\left(\frac{p(F)}{p(F,T)}\right) \\
&\quad + p(T,F)\log_2\left(\frac{p(T)}{p(T,F)}\right) + p(T,T)\log_2\left(\frac{p(T)}{p(T,T)}\right) \\
&= \frac{2}{10}\log_2\left(\frac{\frac{6}{10}}{\frac{2}{10}}\right) + \frac{4}{10}\log_2\left(\frac{\frac{6}{10}}{\frac{4}{10}}\right) \\
&\quad + \frac{3}{10}\log_2\left(\frac{\frac{4}{10}}{\frac{3}{10}}\right) + \frac{1}{10}\log_2\left(\frac{\frac{4}{10}}{\frac{1}{10}}\right) \\
&= 0.875449
\end{aligned}
$$

(c) Find mutual information $I(X_1, Y)$ and $I(X_2, Y)$ to at least 3 decimal places and determine which one (X_1 or X_2) is more informative. [3pts]

As defined, Mutual Information $= I(X_i, Y) = H(Y) - H(Y|X_i)$. As such, we easily arrive at our answers of

$$
\begin{aligned}
I(X_1, Y) &= 1 - 0.960960 = 0.039040 \\
I(X_2, Y) &= 1 - 0.875449 = 0.124551
\end{aligned}
$$

Since Mutual Information is the reduction in uncertainty, we wish to find the higher value. Therefore, we find $I(X_2, Y)$ is more informative.

(d) Find joint entropy $H(Y, X_3)$ to at least 3 decimal places. [3pts]

We define joint entropy as follows:

$$
H(Y, X) = -\sum_{x\in X,\ y\in Y} P(X = x, Y = y) log_2 P(X = x, Y = y)
$$

We recognize we have two cases for Y: $Y = F,\ Y = T$ and two cases for X_3: $X_3 = F,\ X_3 = T$. Therefore, we have the following:

$$
\begin{aligned}
H(Y, X_3) &= \frac{4}{10}log_2(\frac{4}{10}) + \frac{2}{10}log_2(\frac{2}{10}) + \frac{1}{10}log_2(\frac{1}{10}) + \frac{3}{10}log_2(\frac{3}{10}) \\
&= 1.846439
\end{aligned}
$$

5.2 Entropy Proofs [10pts]

(a) Write the discrete case mathematical definition for $H(X|Y)$ and $H(X)$. [3pts]

We will define $H(X)$ and $H(X|Y)$ for discrete random variables below.

$$H(X) = -\sum_{k=1}^{k} P(X = k) log_2(P(X = k))$$

$$H(X|Y) = \sum_{y \in Y} P(Y = y) H(X|Y = y)$$

(b) **Using the mathematical definition of $H(X)$ and $H(X|Y)$ from part (a)**, prove that $I(X,Y) = 0$ if X and Y are statistically independent. (Note: you must provide a mathematical proof and cannot use the visualization shown in class found here. You may use any theorem/ proof from the slides without having to re-prove it). [7pts]

Start from: $I(X,Y) = H(X) - H(X|Y)$

Using $H(X|Y)$, we have the following:

$$H(X|Y = y) = -\sum_{k=1}^{k} P(X = k|Y = y) log_2(P(X = k|Y = y))$$

If X and Y are statistically independent, then we know $P(X = x|Y = y) = P(X = x)$.

Therefore, we can simplify the previous equation to obtain the following:

$$H(X|Y = y) = -\sum_{k=1}^{k} P(X = k) log_2(P(X = k)) = H(X)$$

We have proven $H(X|Y = y) = H(X)$. Therefore, we may substitute back into the original equation of $H(X|Y)$.

$$H(X|Y) = \sum_{y \in Y} P(y) H(X|Y = y) = \sum_{y \in Y} P(Y = y) H(X) = H(X) \sum_{y \in Y} P(Y = y)$$

By definition, we know the sum of probabilities of $Y = y$ is 1. Therefore, we have the following:

$$H(X|Y) = H(X) \times 1 = H(X)$$

Now, we return to the starting point of $I(X,Y) = H(X) - H(X,Y)$.

$$I(X,Y) = H(X) - H(X,Y) = H(X) - H(X) = 0$$

Therefore, we have shown that if X and Y are statistically independent, then $I(X,Y) = 0$.

6 Ethical Implications on Decision-Making [5 pts]

Real-world Implications

Loan eligibility determines who can receive a loan, typically based on financial history and demographics. It is a difficult problem, and often uses algorithms to make loan decisions. Often, this can result in reinforcing inequality and bias [1].

Suppose we're using a matrix to represent the attributes of individuals for loan approval. Each attribute (like income, credit score, years of employment, etc.) constitutes a column in our matrix. Here's a hypothetical toy example:

	Annual Income	Debt-to-Income Ratio	Employment History (years)	Credit Score
Candidate 1	50,000	0.2	5	700
Candidate 2	51,000	0.21	5.1	710
Candidate 3	45,000	0.19	4.9	690
Candidate 4	100,000	0.05	10	780

One algorithm used to predict credit score is linear regression, formulated as $\mathbf{y} = \mathbf{x}\mathbf{A}$. $\mathbf{y}$ are the target variables, $\mathbf{x}$ are the input features, and $\mathbf{A}$ is a matrix trained with an existing dataset. Training data $(\mathbf{x_D}, \mathbf{y_D})$ are taken from the training dataset D, $(\mathbf{x_D}, \mathbf{y_D}) \in D$. If $\mathbf{x_D}$ is linearly independent, $\mathbf{A}$ can be trained by simply inverting $\mathbf{x_D}$:

$$\begin{aligned} \mathbf{y_D} &= \mathbf{x_D}\mathbf{A} \\ \mathbf{x_D}^{-1}\mathbf{y_D} &= \mathbf{A} \end{aligned}$$

The original equation can be rewritten as:

$$\begin{aligned} \mathbf{y} &= \mathbf{x}\mathbf{A} \\ &= \mathbf{x}\mathbf{x_D}^{-1}\mathbf{y_D} \end{aligned}$$

Problems arise when the training data is close to linearly dependent. Recall that one way to invert a matrix is $\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})}\text{adj}(\mathbf{A})$. As $\mathbf{A}$ becomes more linearly dependent and $\det(\mathbf{A}) \to 0$, $||\mathbf{A}^{-1}||$ can become so large it causes numerical errors. Rewriting the original equation:

$$\begin{aligned} \mathbf{y} &= \mathbf{x}\mathbf{x_D}^{-1}\mathbf{y_D} \\ &= \frac{1}{\det(\mathbf{x_D})}\mathbf{x}\,\text{adj}(\mathbf{x_D})\mathbf{y_D} \end{aligned}$$

The errors caused by $\det(\mathbf{x_D}) \to 0$ propagate to $\mathbf{y}$, causing predictions to be wildly inaccurate anywhere outside of the original training set.

Practical Implications

1. Instability: With a small determinant, minor variations in the attributes can lead to significant variations in the results. So, a small difference in income might result in a disproportionate change in loan eligibility.

2. Poor Generalization: If the matrix is based on data with limited variation (like our small community example), it's essentially trained on a very narrow subset of potential applicants. If someone from outside this narrow subset applies (e.g., a person with a 2-year employment but a $70,000 income), the system may not process their application fairly or accurately because it's unfamiliar with such profiles.

Given that a matrix used for determining loan approvals has a determinant close to zero due to limited variation in applicants' attributes:
Which of the following implications might this have on the decision-making process? Choose as all options that apply. Use "textbf{}" to select your answer.

A) It ensures a more uniform scoring system since most applicants have similar attributes.

B) It can lead to unpredictable scores, where tiny variations in attributes yield vastly different outcomes.

C) The system is more resilient to errors because of the limited attribute variation.

D) It might not generalize well to broader populations, potentially leading to biases when applied to more diverse applicant groups.

Answer Here:

7 Programming [5 pts]

See the Programming subfolder in Canvas.

8 Bonus for All [8%]

(a) Let X, Y be **two statistically independent** $N(0,1)$ random variables, and P, Q be random variables defined as:

$$\begin{aligned} P &= 2X + 5XY^2 \\ Q &= X \end{aligned}$$

Calculate the variance $Var(P+Q)$. *(This question may take substantial work to support, e.g. 25 to 30 lines)* [4%]

HINT: The following equality may be useful: $Var(XY) = E[X^2Y^2] - [E(XY)]^2$
HINT: $E[Y^4] = \int_{-\infty}^{\infty} y^4 f_Y(y)dy$ where $f_Y(y)$ is the probability density function of Y (Wolfram alpha calculator or other similar calculators can be used)
HINT: $Var(P+Q) = Var(P) + Var(Q) + 2Cov(P,Q)$ may be a good starting point.

We begin by taking the advice of the TA's and starting with the third hint.

We begin by calculating $Var(P)$.

$$\begin{aligned} Var(P) &= Var(2X + 5XY^2) \\ &= Var(2X) + Var(5XY^2) \\ &= 2^2Var(X) + 5^2Var(XY^2) \\ &= 4(1) + 25(E[X^2Y^4] - (E[XY^2])^2 \\ &= 4 + 25(E[X^2]E[Y^4] - (E[X]E[Y^2])^2 \\ &= 4 + 25((1)(\int_{-\infty}^{\infty} y^4(\frac{1}{\sqrt{2\pi}})e^{-\frac{y^2}{2}}dy) - ((0)E[Y^2])^2 \\ &= 4 + 25(3-0) \\ &= 79 \end{aligned}$$

Next, we will calculate $Var(Q)$.

$$\begin{aligned} Var(Q) &= Var(X) \\ &= 1 \end{aligned}$$

Finally, we will calculate $Cov(P,Q)$.

$$\begin{aligned} Cov(P,Q) &= Cov(2X + 5XY^2, X) \\ &= Cov(2X, X) + Cov(5XY^2, X) \\ &= 2Var(X) + 5E[X^2Y^2] - 5E[XY^2]E[X] \\ &= 2(1) + 5E[X^2Y^2] - 0 \\ &= 2 + 5E[X^2]E[Y^2] \\ &= 2 + 5(1) \\ &= 7 \end{aligned}$$

Therefore, we have the following:

$$\begin{aligned} Var(P+Q) &= Var(P) + Var(Q) + 2Cov(P,Q) \\ &= 79 + 1 + 2(7) \\ &= 94 \end{aligned}$$

(b) Suppose that X and Y have joint pdf given by:

$$f_{X,Y}(x,y) = \begin{cases} \dfrac{1}{24}xe^{-\frac{1}{3}y} & 0 \leq x \leq 4, y \geq 0 \\ 0 & otherwise \end{cases}$$

What are the marginal probability density functions for X and Y? *(It is possible to thoroughly support your answer to this question in 8 to 10 lines)* [2%]

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x,y)dy \\ &= \int_0^{\infty} \frac{1}{24}xe^{-\frac{1}{3}y}dy \\ &= \frac{x}{24}\int_0^{\infty} e^{-\frac{1}{3}y}dy \\ &= \frac{x}{24}[-3e^{-\frac{1}{3}y}]_0^{\infty} \\ &= \frac{x}{8} \ for\ 0 \leq x \leq 4 \end{aligned}$$

$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} f_{X,Y}(x,y)dx \\ &= \int_0^{4} \frac{1}{24}xe^{-\frac{1}{3}y}dy \\ &= \frac{e^{-\frac{1}{3}y}}{24}\int_0^{4} xdx \\ &= \frac{e^{-\frac{1}{3}y}}{24}[\frac{x^2}{2}]_0^4 \\ &= \frac{1}{3}e^{-\frac{1}{3}y} \ for\ y \geq 0 \end{aligned}$$

(c) A person decides to toss a biased coin with $P(heads) = 0.25$ repeatedly until he gets a head. He will make at most 6 tosses. Let the random variable Y denote the number of heads. Find the probability distribution of Y. Then, find the variance of Y. Round your answer to 3 decimal places. *(It is possible to thoroughly support your answer to this question in 5 to 10 lines)* [2%]

We have the random variable Y denote the number of heads. According to the problem, there are two situations: $Y = 0$ (no heads) or $Y = 1$ (one head). We calculate $P(Y = 0) = (0.75)^6 = 0.1780$.

Similarly, $P(Y = 1) = 1 - P(Y = 0) = 1 - 0.1780 = 0.8220$.Therefore, we have arrived at the probability distribution for Y:

$$P(Y = 0) = 0.1780 \ and \ P(Y = 1) = 0.8220$$

Now, we will calculate the Variance.

$$\begin{aligned} Var(Y) &= E[Y^2] - (E[Y])^2 \\ &= (0^2P(Y = 0) + 1^2P(Y = 1)) - (0P(Y = 0) + 1P(Y = 1))^2 \\ &= (0 + 0.8220) - (0 + 0.8220)^2 \\ &= 0.8220 - 0.8220^2 \\ &= 0.1463 \end{aligned}$$

Therefore, we have arrived at our answer.

References

[1] Cathy O'Neil. *Weapons of Math Destruction*. Penguin Books, 2017.